

Allen Mao

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Skills

CAD: Siemens NX, CATIA V5, SolidWorks

Analysis: Thermal Desktop, Abaqus, LS-DYNA, STAR-CCM+

Experience

Senior AI Engineer, Ansys (part of Synopsys) – Sunnyvale, CA (Remote) Jan 2024 – Present

- Architected a cloud-native data ingestion platform (Flask, Pulumi, AWS EKS, VTK) to automate mesh and metadata extraction from physics simulations, enabling scalable data prep for ML training.
- Defined “canonical data” standards to decouple fragile ML logic from raw CAD/simulation formats, implementing robust CPU-side validation to prevent expensive failures during downstream GPU training.
- Optimized geometric processing performance by refactoring pipelines from PyVista to native VTK, achieving a 25% reduction in memory footprint and enabling support for large-scale **VTXHDF** datasets.
- Advanced the open-source ecosystem by upstreaming patches to lasso-python and navigating **VTK C++ source** and **CMake** builds to identify and report core library bugs to maintainers.
- Led cross-functional initiatives with Fluent, Enight, and LSTC teams to integrate native SDKs, eliminating manual data conversion and streamlining simulation-to-ML customer workflows.

R&D Engineer II, CTO Office, Ansys – San Jose, CA (Remote) Sep 2022 – Dec 2023

- Refactored a proprietary, library-agnostic ML framework into a modular, object-oriented architecture, resulting in widespread adoption by internal researchers who previously relied on raw PyTorch and TensorFlow.
- Architected a multi-modal web platform (Go, Angular, Python) utilizing AWS SageMaker and Nvidia Triton to expose physics-informed ML models, enabling engineers to run ML-based physics simulation.
- Engineered a scalable research environment by deploying Jupyter Enterprise Gateway on EKS with isolated kernels, eliminating cross-library dependency conflicts and reducing environment setup time for the R&D team.

Software Development Engineering Intern, Amazon – Portland, OR (Remote) May 2021 – Aug 2021

- Modernized a legacy vendor search system by conducting stakeholder research and log analysis to architect a responsive Vue.JS and Spring application.
- Implemented a Backend-for-Frontend (BFF) pattern to aggregate data from internal Vendor APIs, streamlining listings management for enterprise-scale partners.

Software Engineering Intern, NASA Kennedy Space Center – Merritt Island, FL (Remote) Aug 2020 – Dec 2020

- Engineered system and acceptance tests for the **Artemis** Launch Control System GUI, ensuring mission-critical software reliability for the Space Launch System (SLS) ground operations.

Projects

Software Metadata Extraction Framework (SOMEF) github.com/KnowledgeCaptureAndDiscovery/somef

- The more work experience you have, the less relevant outside-work projects tend to become
- If you have something that really stands out, consider listing it here

Project Title [project.com](#)

- Only list real projects, not mandatory school projects or trivial tutorial projects found online
- Something that someone uses to solve a problem
- Something that has users (can be just you, as long as you use it often) and is actively maintained and not just rotting in a GitHub repo, never to see a PR for the rest of its life

Education

University of California, Berkeley – BA in Computer Science

May 2022